

ANIMESH SHUKLA

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OBJECTIVE

An aspiring Electronics and Communication Engineering student with a strong interest in both hardware systems and software development, eager to apply analytical, programming, and circuit design skills through internships and projects.

EDUCATION

Sardar Vallabhbhai National Institute of Technology (SVNIT)	Surat, India
<i>B.Tech – Electronics and Communication Engineering CGPA: 7.81/10 (3rd Semester)</i>	<i>2024 – Present</i>
Class XII – CBSE Percentage: 88.4%	Class X – CBSE Percentage: 90.2%

TECHNICAL SKILLS

Languages: C, Embedded C, Python, HTML, CSS
Software / Tools: Arduino IDE, Keil uVision, LTspice, MATLAB, VS Code, MS Office
Operating Systems: Windows, Linux, Android
Hardware: Arduino UNO/Nano, ESP32, MPU6050, IR/Ultrasonic Sensors, Encoder Motors, L298N, Breadboard, Soldering Iron
Concepts: Embedded Systems, Circuit Design, PID Control, Sensor Fusion, Inverse Kinematics, Odometry

PROJECTS

Autonomous Go-to-Goal Mecanum Drive Robot 2025

- Designed and implemented a Go-to-Goal navigation algorithm using encoder-based odometry and MPU6050 gyroscope data for real-time position and heading estimation.
- Derived inverse kinematics for the mecanum wheel configuration and implemented PID control loops for stable, accurate omnidirectional motion toward target coordinates.
- Diagnosed and resolved sensor noise, voltage instability, and motor inconsistencies; programmed the complete control stack in Embedded C on Arduino.

International Rover Challenge 2026 – Rover Build 2026

- Selected to represent SVNIT at IRC 2026, Udupi – an international inter-collegiate competition simulating Mars surface exploration tasks.
- Contributing to electronics and embedded systems subsystems including motor control, power distribution, and sensor integration under strict weight and power constraints.
- Applying iterative prototyping and cross-functional team coordination through regular design reviews and system testing cycles.

Line Follower Robot 2024

- Built an autonomous line-following robot in a 4-member team using Arduino UNO, IR sensor array, and L298N motor driver with real-time threshold-based path detection.
- Calibrated sensor sensitivity and tuned motor speed, securing 1st Runner-Up at the college-level robotics competition.

POSITIONS OF RESPONSIBILITY

Co-ordinator – Sparsh 2025, Social and Cultural Fest, SVNIT

- Managed end-to-end event planning, volunteer coordination, and logistics; served as the key communication bridge between the organizing committee and participants.

Executive (Discipline) – Mindbend 2025, Technical Fest, SVNIT

- Oversaw discipline and orderly conduct across multiple technical event venues, coordinating with event heads during the annual technical festival.

Active Member – Unnat Bharat Abhiyan, Government of India 2024 – Present

- Contributing to rural development and community outreach programs, applying student-driven engineering perspectives to grassroots-level initiatives.

Volunteer & Member – SPIC MACAY | Think India 2024 – Present

- Volunteered at classical arts and cultural events under SPIC MACAY; engaged in national policy discussions and student leadership programs through Think India.